



SCHOOL OF
SCIENCE &
TECHNOLOGY

Capstone Project Proposals

Master in Business Analytics and Data Science
Master in Computer Science and Business Technology

Academic Year 2025/2026

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Introduction

This document provides an overview of the capstone project proposals offered by faculty members for students enrolled in the **Master in Business Analytics and Data Science** (MBADS) and the **Master in Computer Science and Business Technology** (MCSBT). Each proposal includes a brief description of the topic, methodological focus, and potential application areas.

Supervisor: Bissan Ghaddar (bghaddar@faculty.ie.edu)

Optimization of Carbon-Aware AI Workload Scheduling

This project focuses on optimization of shifting AI training workloads both spatially (across geographically distributed data centers) and temporally (across time periods) in order to reduce carbon emissions while maintaining acceptable training performance and service constraints.

The work will:

- Develop mathematical optimization models that account for data center capacities, workload latency, transfer costs, and time-varying carbon intensity of electricity.
- Use real carbon-intensity data (e.g., Electricity Maps) and real workload traces.
- Quantify potential emissions reductions achievable through carbon-aware workload scheduling.

Supervisor: Nacho Molina (nacho.molina@ie.edu)

Project 1: Graph-Based Concept Diffusion Metrics to Evaluate Research Impact

This project develops a data-driven framework to measure scientific impact by tracking how concepts emerge, spread, and persist across the research ecosystem. Using large-scale publication metadata (titles and abstracts, timestamps, venues, authorship) and modern NLP embeddings, students will build semantic networks of papers and open-vocabulary concept graphs to quantify idea diffusion over time.

The main outcomes include:

- Alternative impact indicators at the level of papers, researchers, journals, institutions, and universities.
- Metrics based on the breadth, speed, and longevity with which concepts propagate across fields and communities.
- A reproducible analytics pipeline and interactive dashboard that compare traditional metrics (citations, h-index, journal impact factors) with concept-diffusion impact scores.

Project 2: The Impossibility of Modeling Ourselves

This project studies how predictive models in business analytics and economic forecasting can change the very systems they are intended to model. Using an agent-based simulation of an economic system, it analyzes how firms, consumers, or investors adapt their decisions when they adopt a successful forecasting model.

Key components:

- Simulation of model adoption and its impact on market dynamics.
- Quantification of forecast decay as adoption spreads.
- Evaluation of whether retraining or model updating can restore predictive power.

Supervisor: Manuele Leonelli (manuele.leonelli@ie.edu)

Project 1: Bayesian Hierarchical Models for Global and Regional Indicators

This project applies Bayesian hierarchical modeling to structured cross-country datasets (e.g., Human Rights indices, Travel & Tourism Development Index, European Innovation Scoreboard, Global Food Security Index). These data sets share a multilevel structure, with countries nested within regions and observed across years.

The project will:

- Build Bayesian multilevel models that capture country-level heterogeneity with partial pooling within regions.
- Include varying intercepts, varying slopes, time effects, or cross-classified structures as appropriate.
- Emphasize interpretation of shrinkage effects, regional variability, and uncertainty in rankings and comparisons.

Project 2: Bayesian Sports Modeling (Dynamic Elo and Hierarchical Models)

This project studies Bayesian approaches to sports performance modeling, combining methodological understanding with structured application. Two possible directions are Bayesian dynamic Elo models or hierarchical models for sports outcomes (e.g., goals, sets, points).

Students will:

- Specify probabilistic models with hierarchical priors for latent strength parameters.
- Use real sports datasets (e.g., football matches, tennis matches, volleyball rallies).
- Evaluate model performance using predictive accuracy and posterior predictive checks.

Project 3: Hierarchical Extreme Value Models

This project investigates Bayesian hierarchical extreme value models using Generalized Extreme Value (GEV) or Generalized Pareto (GPD) distributions with partial pooling across groups.

Example applications include:

- Extreme financial returns across sectors or crisis versus non-crisis periods.
- World-record performances or extreme scoring events in sports.

- Engineering extremes such as skyscraper heights or infrastructure records.

The focus is on modeling tail behavior, quantifying uncertainty in return levels, and comparing group-level tail risk using posterior inference and predictive evaluation.

Supervisor: Andrea Cremaschi (acremaschi@faculty.ie.edu)

Bayesian Neural Networks Based on Modern Architectures

This project proposes the implementation of Bayesian neural networks based on modern architectures such as Kolmogorov–Arnold Networks. The goal is to implement these models in a rigorous Bayesian framework, studying the effect of different prior choices and incorporating existing information when possible.

The work will:

- Develop Bayesian versions of state-of-the-art neural network architectures.
- Analyze interpretability gains from prior–posterior relationships.
- Provide uncertainty quantification and apply the models to benchmark datasets, with possible extensions to more complex Bayesian structures.

Supervisor: Javier Muro (javier.muro@ie.edu)

Geospatial Data Analytics and Applications

The research ideas in this area must involve geospatial data (satellite imagery and/or other mapping products). Ideally, the student will apply time series analysis and/or machine learning to solve a concrete problem.

Note: This proposal defines a research area rather than a single fixed project. The specific topic and research question will be defined collaboratively between the student and the supervisor during the initial meetings.

Example application areas include:

- Risk reduction and disaster response.
- Nature conservation.
- Renewable energy.
- Food security and agriculture.
- Maritime piracy (illegal ship identification).
- Spatial consequences of armed conflicts.

Alternatively, projects involving the development of applications for geospatial data visualization will also be considered.

Supervisor: Rubén Sánchez García (rsanchezgarcia@faculty.ie.edu)

Project 1: Neural Architectures for Fourier-Space Data in Cryo-EM

This project explores novel neural architectures for Fourier-space data processing with applications in cryo-electron microscopy (cryo-EM). It builds on recent advances such as Fourier Image Transformers and neural operator frameworks to design models that explicitly operate in frequency space.

Key objectives:

- Develop architectures that capture long-range frequency correlations and exploit physical symmetries in cryo-EM data.
- Ensure computational efficiency at high resolution and robustness to noise and missing data.
- Conduct theoretical analysis of spectral representation properties and empirical evaluation on large-scale 2D and 3D cryo-EM tasks.

Project 2: DETR-Inspired Architectures for Atom Detection in Cryo-EM 3D Maps

This project adapts DETR-like transformer-based object detection architectures to the problem of atom detection in 3D cryo-EM density maps. The models treat atomic positions as a set prediction problem with attention-based mechanisms.

The project will:

- Design models that take 3D density volumes as input and predict sets of atomic coordinates.
- Integrate biophysical priors (e.g., bond lengths, valence, stereochemistry) into the architecture or loss functions.
- Compare performance against U-Net-3D-based baselines on benchmark cryo-EM datasets in terms of accuracy, robustness, and scalability.

Supervisor: Luciano Dybala (luciano.dybala@ie.edu)

Project 1: AI Interpretability

Students will experiment with techniques for visualizing and explaining the inner workings of deep neural networks, and potentially biological neural networks using brain data. Projects may involve implementing a visualization tool or library, as well as conducting research on current interpretability methods and developing new ones.

Proficiency with PyTorch is a prerequisite for this project.

Project 2: Dimensionality Reduction and Data Visualization

This project focuses on nonlinear dimensionality reduction methods for summarizing data and creating informative visualizations (e.g., t-SNE, UMAP, diffusion maps). Possible directions include:

- Proposing modifications or improvements to existing dimensionality reduction methods.
- Developing novel metrics to evaluate the quality of low-dimensional embeddings.
- Applying these techniques to complex, high-dimensional datasets relevant to business analytics or computer science.

Supervisor: Dae-Jin Lee (dae-jin.lee@ie.edu)

Exact Cross-Frequency Aggregation in Mixed-Frequency Time Series

This project develops a unified statistical framework for mixed-frequency time series in which low-frequency outcomes (e.g., monthly, quarterly) depend on covariates observed at higher frequencies (e.g., daily, weekly). Standard models such as ARIMAX and MIDAS either ignore the exact aggregation constraint or rely on restrictive lag specifications; this project replaces ad hoc pre-aggregation with a principled composite link state-space (CLM-SS) formulation.

The project introduces a cross-frequency time series framework using a composite link matrix for exact aggregation, flexible trends, and ARIMA errors in state-space form. Methodological aims include identifiability analysis, penalized likelihood estimation, and comparison with MIDAS and ARIMAX benchmarks.

Empirical applications will focus on publicly available economic and energy datasets, such as monthly inflation with daily oil prices and hourly electricity demand with daily temperature and monthly fuel costs, using freely accessible sources (e.g., FRED, ENTSO-E, Open Power System Data). Deliverables include a documented Python/R codebase, a simulation study, empirical results, and a written report targeting an applied econometrics or forecasting journal.